Gesture-Based Assistive Communication Using TinyML and FSK for Real-Time Embedded Systems

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***Abstract*—This paper presents a wearable assistive communication system that translates hand gestures into text and audible speech in real time. The system uses an MPU6050 motion sensor to capture hand movements and an ESP32 microcontroller to process the sensor data. A lightweight TinyML model is deployed directly on the ESP32 for on-device, low-latency, and offline gesture recognition. Once a gesture is identified, it is converted into predefined text and speechfor immediate user feedback. In addition, the recognized message is encoded into binary form and demonstrated using Frequency Shift Keying (FSK) modulation in MATLAB for communication signal analysis. The innovation of the proposed system lies in the integration of embedded TinyML inference, assistive gesture-to-speech output, and FSK-based communication demonstration within a single platform. By combining wearable sensing, edge intelligence, and real-time speech generation, the system provides a low-cost, portable, offline, and accessible solution for individuals with speech or hearing impairments.**

Keywords: TinyML, Gesture Recognition, Assistive Communication, Gesture-to-Speech, ESP32, MPU6050, Embedded Systems, FSK Modulation, Edge AI.

I. Introduction

The rapid advancement in wearable technology and embedded sensor systems has led to new opportunities for improving human-computer interaction. Among these emerging technologies, gesture-based interaction systems have gained significant popularity due to their ability to offer users a more natural way of interacting with computing systems. A prime example of this trend is the Gesture-Based Assistive Communication System, which utilizes motion sensors, TinyML, and FSK communication protocols for recognizing hand gestures and converting them into meaningful information in the form of text or speech.

The main component of this communication system is a motion sensor known as MPU6050, which consists of both an accelerometer and a gyroscope for accurately recognizing hand gestures. The sensor is embedded in a wearable device for recognizing hand gestures performed by users. The ESP32 microcontroller is used for processing sensor information, utilizing TinyML for recognizing gestures, and sending information through a Frequency Shift Keying (FSK) protocol for communication with a receiver. At the receiving end, a Python-based interface is used for visualizing gestures and converting them into speech for communication by users who suffer from speech or motor disorders.

The importance of the system is based on the fact that it is capable of providing an affordable, portable, and real-time assistive communication system that can be used entirely offline without the need for cloud-based computation. This project is based on the integration of sensor technology, machine learning, and digital communication, providing a platform that can be used to improve the life of people with disabilities.

This study is based on the exploration of the technological aspects of the system, including the sensor integration, sensor processing, gesture recognition using TinyML, and FSK-based communication. In addition, the study is based on the exploration of the potential applications of the system in the real world, including the importance of wearable devices in the real world and the significance of innovation in wearable technology.

II. Literature Review

Over the past few years, there has been substantial research in the development of wearable technology for gesture recognition, especially for assistive communication purposes. In the initial stages of the study, there was the usage of gyroscopes and accelerometers for the detection of hand movements. This was followed by the analysis of the data collected from the sensors to recognize the gestures. Some of the examples of the usage of gesture recognition include the development of data gloves and the Wii Remote. However, there have been challenges in the accuracy of the results obtained from the sensors.

The development of sensor technology has been one of the most important factors in the development of the precision of gesture recognition systems. In recent times, the MPU6050 sensor has been widely used for the purpose of gesture recognition. It is a combination of the accelerometer and the gyroscope sensor. It has been observed that the usage of sensor fusion techniques and the development of proper programming algorithms have resulted in the precision of the results obtained from the sensors.

At the same time, the developments in machine learning, especially lightweight machine learning, have enabled the shift from the offline post-processing of gestures to real-time computation. TinyML has enabled the ESP32 microcontroller to perform complex computations on the embedded system itself, eliminating the constant need for connectivity to the cloud. Various recent studies have proven the effectiveness of the integration of TinyML with wearable devices not only

in terms of real-time feedback but also in terms of the cost-effectiveness of the assistive system.

Recent studies have highlighted the significance of real-time feedback in the development of assistive systems based on gestures. Mobile applications have been designed to provide real-time feedback to the user in the form of visual and auditory feedback. Various studies have proven the effectiveness of the combination of wearable devices with interactive software in the development of assistive systems for people with disabilities. It has been proven that gestures can be translated into text or speech through the combination of wearable devices with interactive software.

Overall, these developments highlight the promise of sensor technology, machine learning, and interactive technology in delivering advanced assistive technology solutions. Leveraging previous research in motion sensing, embedded technology, and feedback, gesture recognition technology can now deliver more accurate, user-friendly, and pragmatic solutions for facilitating communication for those with impairments.

III. Methodology

A. System Architecture

The gesture-to-speech communication system, as proposed, utilizes the MPU6050 gesture sensor, ESP32 microcontroller, and MATLAB processing environment for real-time communication. The MPU6050 gesture sensor utilizes its in-built accelerometer and gyroscope to detect hand gestures based on its acceleration and angular velocity. The ESP32 microcontroller is connected to the MPU6050 gesture sensor using the I2C communication protocol and continuously reads the data from the hand gestures.

The ESP32 microcontroller is programmed using the Arduino Integrated Development Environment and is used for initial data processing, such as noise removal and normalization. A TinyML gesture recognition model is also integrated with the ESP32 microcontroller for real-time gesture recognition. Once a valid gesture is recognized by the model, the gesture information is sent to the computer system using serial communication.

The gesture information is processed in the MATLAB environment and converted to a predefined text message based on the gesture recognition. The text message is converted to a speech signal using the Windows Speech Synthesizer interface. In addition, the text message is converted to a binary format and modulated using the Frequency Shift Keying (FSK) technique. MATLAB tools, such as time domain and spectrogram plots, are used for analyzing the gesture and communication signals.

B. Block Diagram

The block diagram of the proposed system, Gesture-Based Assistive Communication Using TinyML FSK, demonstrates an organized way of combining hardware and software components for hand gestures, speech generation, and communication signal processing. The heart of the system is an ESP32 microcontroller, which acts as the brain of the system. The microcontroller is connected to an MPU6050 sensor, which is fixed on a band or glove and acts as a motion sensor for precise detection of hand movements. The MPU6050 sensor contains an accelerometer and gyroscope, which are used to detect both linear and rotational movements of the hand.

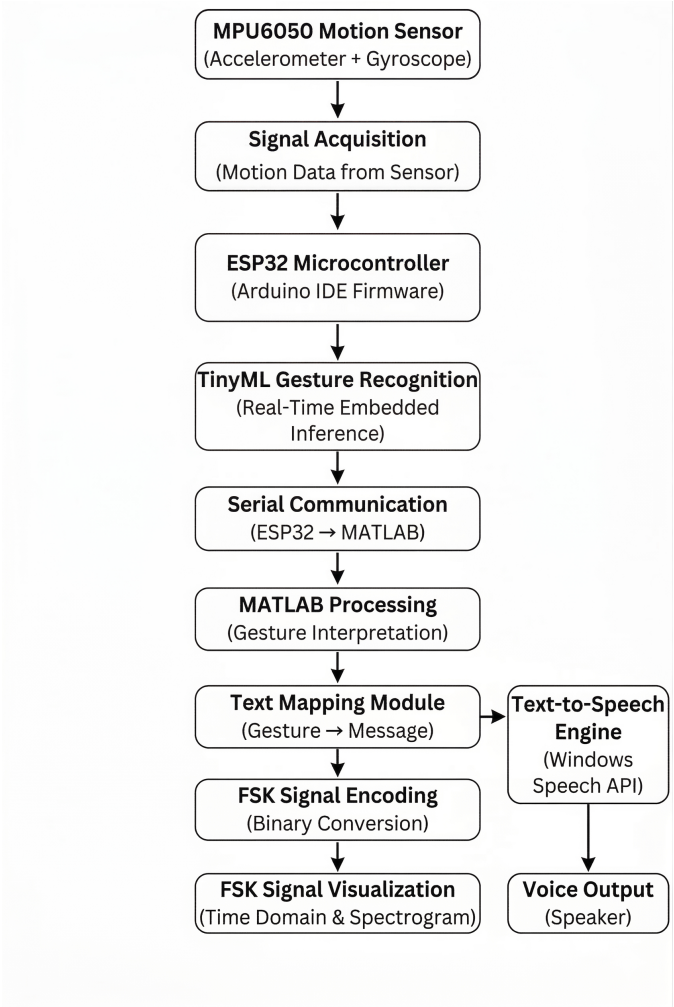

Fig. 1. System Architecture of Gesture-to-Speech Communication

The ESP32 microcontroller contains firmware that uses the Arduino IDE for acquiring, preprocessing, and recognizing hand gestures. The ESP32 microcontroller reads raw data from the MPU6050 sensor using the I2C communication protocol and filters, normalizes, and calibrates the data to obtain accurate readings. The ESP32 microcontroller contains a TinyML-based module that classifies predefined hand gestures. The system uses the ESP32 microcontroller to perform offline gesture recognition without relying on cloud services.

Once a gesture is recognized, the ESP32 sends the classified gesture data to a computer system through serial communication. MATLAB is used for signal analysis, waveform visualization, and gesture validation. The recognized gesture is mapped to a predefined text message, which is then converted into audible speech using MATLAB and the Windows Speech Synthesizer through a text-to-speech engine.

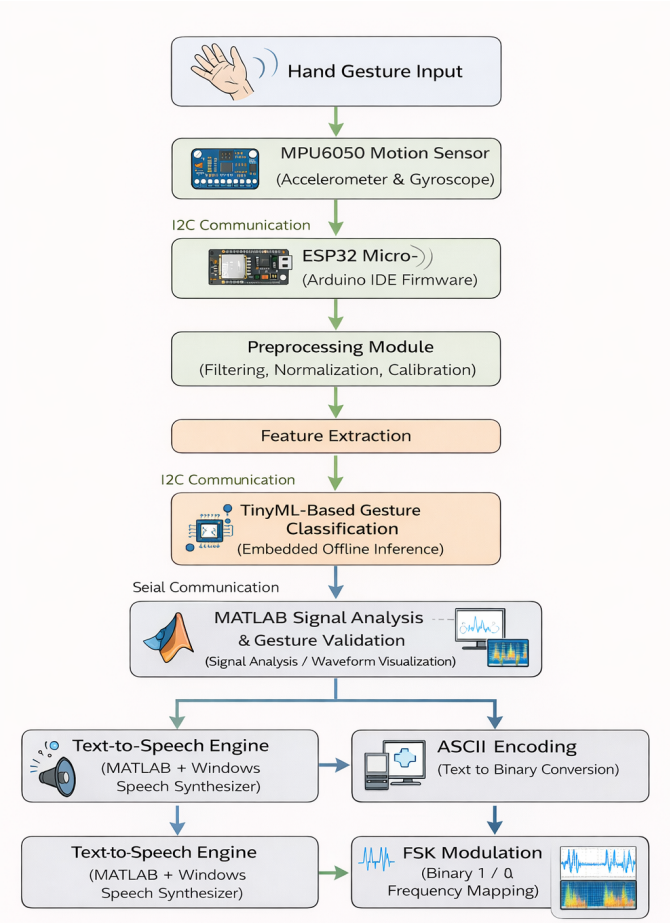

Fig. 2. Block Diagram Of Gesture-To-Speech Communication

In addition to speech generation, the system demonstrates digital communication using Frequency Shift Keying (FSK) modulation. The recognized text is converted into binary form using ASCII encoding, and the binary data is modulated using two distinct carrier frequencies representing binary '1' and binary '0'. MATLAB generates the corresponding FSK waveform and enables spectrogram analysis to validate the modulation process. The communication protocols used in the system include I2C for sensor interfacing and serial communication for data transfer. The system is designed to be portable, efficient, and suitable for real-time assistive communication applications.

The methodology for developing the Gesture-Based Assistive Communication Using TinyML & FSK system includes hardware integration, embedded processing, signal analysis, gesture recognition, digital communication, and testing.

Firstly, the hardware design phase includes hardware integration, where an ESP32 microcontroller and an MPU6050 accelerometer and gyroscope sensor are integrated and connected to a wearable band or glove. The MPU6050 sensor collects real-time data on hand movement and orientation, and the ESP32 acts as the central processing unit. The hardware assembly is designed to be portable, comfortable, and effective using a rechargeable power source.

Secondly, the software development phase includes firmware and software implementation, where the ESP32 and Python-based visualization are integrated. The ESP32, using the Arduino IDE, collects data and preprocesses it using filters, normalization, and feature extraction. The ESP32 uses TinyML, where the hand gestures are classified in real-time, and the identified gesture is converted to text or speech using MATLAB.

Thirdly, the use of MATLAB facilitates signal analysis and waveform visualization of the accelerometer and gyroscope signals. This enhances the reliability and accuracy of the gesture recognition system.

Fourthly, the classified gesture data is then transmitted using a technique called Frequency Shift Keying (FSK) modulation. This technique modulates the signal so that it can be decoded by the receiver. When the signal reaches the receiver, the FSK signal is demodulated and decoded to obtain the gesture data, which is then displayed as text or speech.

Lastly, the system also includes a process of calibrating the sensors to ensure the accuracy of the system. This includes the use of the MPU6050, which is calibrated to reduce offset errors. This enhances the reliability and accuracy of the system.

IV. Results And Discussion

A. Hardware and Software Integration

In the proposed system, gesture sensing, embedded processing, and speech generation are integrated to enable real-time gesture-to-speech communication. The system utilizes an MPU6050 motion sensor connected to an ESP32 microcontroller to capture hand motion data using accelerometer and gyroscope measurements. The ESP32 is programmed using the Arduino IDE and performs initial signal preprocessing, including noise filtering and normalization of the sensor data. A TinyML-based gesture recognition model is deployed on the ESP32 to classify hand gestures in real time directly on the embedded device. Once a gesture is detected, the recognized gesture information is transmitted to the MATLAB environment through serial communication. MATLAB processes the received gesture data and converts the recognized gesture into a predefined text message. The text message is then converted into speech using the Windows Speech Synthesizer interface. In addition, the message is encoded into binary data and modulated using the Frequency Shift Keying (FSK) technique for signal analysis.

B. Gesture Recognition and System Performance

Gesture recognition in the proposed system is performed using a TinyML-based model deployed on the ESP32 microcontroller. The model analyzes motion data obtained from the MPU6050 sensor and classifies the gestures into predefined categories. Experimental observations show that the system can successfully distinguish between active gestures and idle states using the trained TinyML model. The integration of embedded machine learning enables real-time gesture recognition with low computational overhead. Once the gesture is

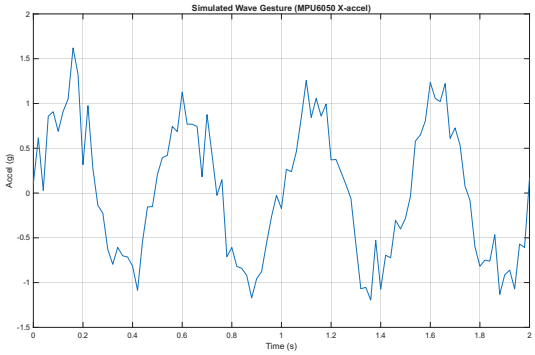

Fig. 3. Simulated Gesture Signal From MPU6050 Accelerometer

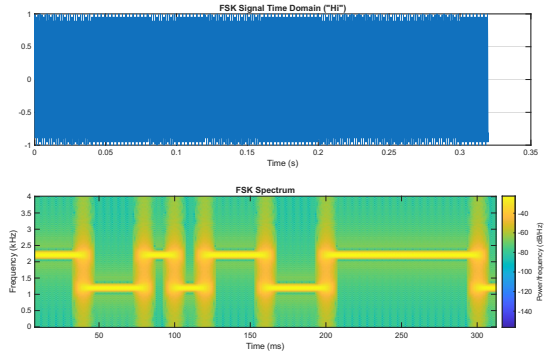

Fig. 4. Time-Domain FSK Waveform and Spectral Analysis

identified, the corresponding message is generated and transmitted for further processing. The generated binary message is modulated using the FSK technique, where two different frequencies represent binary data. The resulting FSK signal clearly demonstrates frequency switching corresponding to the transmitted binary sequence, confirming the correct implementation of the modulation technique.

C. Voice Output and System Interaction

After the gesture is successfully recognized, the corresponding text message is converted into speech using MATLAB's integration with the Windows Speech Synthesizer. The generated speech is played through the system speaker, providing real-time voice feedback to the user. This demonstrates the capability of the system to translate hand gestures into understandable spoken communication. The integration of gesture sensing, TinyML-based recognition, and speech generation highlights the feasibility of developing assistive communication systems for individuals with speech or hearing impairments. The results indicate that the proposed system can effectively detect gestures and generate voice output, enabling intuitive and accessible human–machine interaction.

V. Conclusion

The proposed gesture-to-speech communication system incorporates hardware and software components to recognize hand gestures and convert them into communicable speech in real time. The proposed communication system incorporates an MPU6050 motion sensor to collect hand gesture data using accelerometer and gyroscope readings. The ESP32 microcontroller processes hand gesture data to recognize gestures using a TinyML-based gesture recognition model. The recognized hand gestures are then transmitted to a MATLAB processing environment to convert them into predefined text messages and speech using a text-to-speech engine.

The proposed communication system is not only able to convert hand gestures into communicable speech, but it is also able to demonstrate digital communication capabilities by encoding the generated message using Frequency Shift Keying (FSK) modulation techniques. The experimental results demonstrate that the proposed communication system is able to accurately recognize hand gestures and convert them into communicable speech with minimal delays. The integration of TinyML-based gesture recognition capabilities into the proposed communication system is efficient and accurate.

The proposed communication system demonstrates its capabilities to incorporate efficient communication systems using embedded systems and signal processing techniques to enable efficient communication systems that can greatly improve communication accessibility for people with speech and hearing difficulties.

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